EXAMPLE 6.3

An object is placed at a distance of 20.0 cm from a concave mirror of focal length 15.0 cm.

(a) What distance from the mirror a screen should be placed to get a sharp image?

(b) What is the nature of the image?

Solution

Given, $f = -15$ cm, $u = -20$ cm

(a) Mirror equation, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

Rewriting to find v, $\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$

Substituting for f and u, $\frac{1}{v}=\frac{1}{-15}-\frac{1}{-20}$

$$\frac{1}{v}=\frac{(-20)-(-15)}{300}=\frac{-5}{300}=\frac{-1}{60}$$

$$v=-60.0\ cm$$

The screen is to be placed at distance 60.0 cm to the left of the concave mirror.

(b) Magnification, $m=\frac{h'}{h}=-\frac{v}{u}$

$$m=\frac{h'}{h}=-\frac{(-60)}{(-20)}=-3$$

As the sign of magnification is negative, the image is inverted.

As the magnitude of magnification is 3, the image is enlarged three times.

As the image is formed to the left of the concave mirror, the image is real.

EXAMPLE 6.4

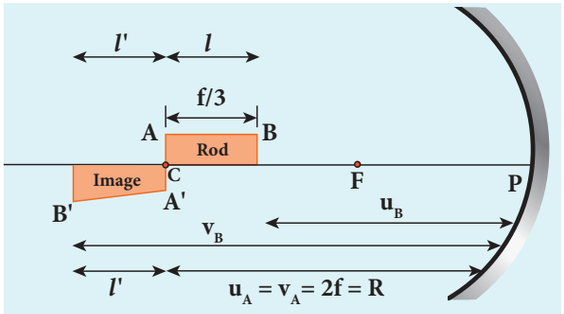
A thin rod of length $f/3$ is placed along the optical axis of a concave mirror of focal length f such that one end of image which is real and elongated just touches the respective end of the rod. Calculate the longitudinal magnification.

Solution

longitudinal magnification $(m_l) = \frac{\text{length of image } (l')}{\text{length of object } (l)}$

Given: length of object, $l = \frac{f}{3}$

For the given condition, the image formation is shown in the figure.



Let, l' be the length of the image, then

$$m_l=\frac{l'}{l}=\frac{l'}{f/3} \text{ (or) } l'=\frac{m_l f}{3}$$

Image of one end coincides with the respective end of object. Thus, the coinciding end must be at centre of curvature.

$$u_B=u_A-\frac{f}{3}=2f-\frac{f}{3}=\frac{5f}{3}$$

$$v_B=u_B+l+l'$$

$$v_B=\frac{5f}{3}+\frac{f}{3}+\frac{m_l f}{3}=\frac{f(6+m_l)}{3}$$

Mirror equation, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$$\frac{1}{-\left(\frac{f(6+m_l)}{3}\right)}+\frac{1}{-\left(\frac{5f}{3}\right)}=\frac{1}{-f}$$

After simplifying,

$$\frac{3}{f(6+m_l)}+\frac{3}{5f}=\frac{1}{f};\frac{3}{(6+m_l)}=\frac{2}{5}$$

$$6+m_l=\frac{15}{2};\ m_l=\frac{15}{2}-6$$

$$m_l=\frac{3}{2}=1.5$$

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